



Semester One Examination, 2019

Question/Answer booklet

**MATHEMATICS
APPLICATIONS
UNIT 1**

**Section One:
Calculator-free**

SOLUTIONS

Student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answer to the specific question asked and to follow any instructions that are specified to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (52 Marks)

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

Evaluate

- (a) $\frac{1}{2}h(a + b)$ when $a = 2.8, b = 4.4, h = 5$ (2 marks)

Solution
$2.5 \times (7.2)$ $= 18$
Specific behaviours
□ indicates correct substitution and calculates either 2.5 or 7.2 ✓ correct value

- (b) $x^2 - 4x + 1$ when $x = \frac{1}{2}$. (2 marks)

Solution
$x^2 - 4x + 1 = \left(\frac{1}{2}\right)^2 - 4\left(\frac{1}{2}\right) + 1$ $= \frac{1}{4} - 2 + 1 = -\frac{3}{4}$
Specific behaviours
□ substitutes correctly ✓ correct value

(c) $b^2 + bc + \sqrt{c}$ when $b = -5$ and $c = 9$.

(2 marks)

Solution
$b^2 + bc + \sqrt{c} = (-5)^2 + (-5)(9) + \sqrt{9} = 25 - 45 + 3$ $= -17$
Specific behaviours
☐ evaluates at least two of the three terms correctly
☒ correct value

Question 2

(6 marks)

One Australian dollar can buy 0.70 US dollars.

(a) How many US dollars can be bought for

(i) 500 Australian dollars?

(1 mark)

Solution
$500 \times 0.70 = \text{US\$}350$
Specific behaviours
☐ correct amount

(ii) 280 Australian dollars?

(1 mark)

Solution
$280 \times 0.70 = \text{US\$}196$
Specific behaviours
☐ correct amount

(b) How many Australian dollars are needed to buy 70 US dollars?

(1 mark)

Solution
$70 \div 0.70 = \text{A\$}100$
Specific behaviours
☐ correct amount

One Australian dollar can buy 10 South African rand.

(c) At an airport a traveller wants to buy 5 000 rand, but the exchange desk also charges them a commission of 3%. Determine the total cost, in Australian dollars, for the traveller to buy this amount of currency. (3 marks)

Solution
$5000 \div 10 = 500$

See next page

$$500 \times \frac{3}{100} = 15$$

$$Total = 500 + 15 = \$515$$

Specific behaviours

- ☐ converts to dollars
- ☒ calculates commission
- ☐ correct total

Question 3

(9 marks)

Four matrices are defined as $A = \begin{bmatrix} 4 & -1 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & -1 & 7 & -2 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 3 & -2 & 0 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & -4 \end{bmatrix}$.

(a) State

(i) which matrix has more rows than columns.

(1 mark)

Solution
<i>Matrix A</i>
Specific behaviours
☐ correct name

(ii) the dimensions of the row matrix.

(1 mark)

Solution
Size of matrix D is 1×2
Specific behaviours
☐ correct size

(b) Calculate $5C + 3I$, where I is the 2×2 identity matrix.

(3 marks)

Solution
$5 \times \begin{bmatrix} 1 & 3 & -2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 15 & -10 & 0 \end{bmatrix}$ $3I = \begin{bmatrix} 3 & 0 & 0 & 3 \end{bmatrix}$ $5C + 3I = \begin{bmatrix} 5 & 15 & -10 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 & 3 \end{bmatrix}$ $= \begin{bmatrix} 8 & 15 & -10 & 3 \end{bmatrix}$
Specific behaviours
☐ correct multiple of C ☒ correct multiple of I ☒ correct sum

(c) Calculate C^2 .

(2 marks)

Solution
$\begin{bmatrix} 1 & 3 & -2 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 3 & -2 & 0 \end{bmatrix}$ $= \begin{bmatrix} -5 & 3 & -2 & -6 \end{bmatrix}$
Specific behaviours
☐ one correct row or column ☒ correct product

(d) It is required to multiply all four matrices together so that the final product is a square matrix with the largest possible number of columns. State the order in which the matrices should be multiplied and state the number of columns in the final product. (2 marks)

Solution
Order: $A \times D \times C \times B$. Product will have 3 columns.
Specific behaviours
☐ correct order ☒ correct number of columns

Question 4

(6 marks)

During the week, casual employees at a recycling plant are paid a wage of \$15 per hour plus \$1.50 for every kg of electronic waste they collect. Over the weekend, both rates are increased by 20%.

A casual worked for 4 hours on Sunday, collecting 30 kg of electronic waste, and then a further 6 hours on Monday when they collected 50 kg of electronic waste.

(a) Determine the hourly pay rate for weekends.

(1 mark)

Solution
$15 \times 0.2 = 3 \Rightarrow \text{New rate is \$18 per hour}$
Specific behaviours
☐ correct rate

(b) Calculate how much the casual earned on Monday.

(2 marks)

Solution
$50 \times 1.50 = 75$
$6 \times 15 = 90$
<i>Total:</i> $90 + 75 = \$165$
Specific behaviours
☐ calculates weight amount
☒ calculates hourly amount and adds to weight amount

(c) Calculate the total earned by the casual over the two days.

(3 marks)

Solution

See next page

$$4 \times 15 + 30 \times 1.5 = 60 + 45 = 105$$

$$105 \times 0.2 = 21$$

$$\textit{Sunday earns } 105 + 21 = \$126$$

$$\textit{Total for 2 days: } 165 + 126 = \$291$$

Specific behaviours

□ Sunday total

✓ increases Sunday total by 20%

✓ correct total for two days

Question 5

(6 marks)

Hockey teams A and B had each played 21 games in a league. Team A had won 11, lost 4 and drawn the remainder of its games. In total, the two teams had won 21 and drawn 14 games.

- (a) Show this information in a 2×3 matrix H , with columns ordered win, lose, draw and rows ordered team A and team B. (2 marks)

Solution
$H = \begin{bmatrix} 11 & 4 & 6 & 10 & 3 & 8 \end{bmatrix}$
Specific behaviours
☐ one row correct ☒ correct matrix

- (b) Each team gains 3 points for a win, 1 point for a draw and no points for a loss. Show this information in matrix P that can be multiplied by matrix H . (1 mark)

Solution
$P = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$
Specific behaviours
☐ correct matrix

- (c) Calculate $H \times P$ and state which team is leading the other in the league. (3 marks)

Solution
$\begin{bmatrix} 11 & 4 & 6 & 10 & 3 & 8 \end{bmatrix} \times \begin{bmatrix} 3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 39 & 38 \end{bmatrix}$ Team A is leading team B.
Specific behaviours
☐ obtains a 2×1 product

- ✓ correct product
- ✓ correct interpretation

Question 6

(7 marks)

An investor held 400, 200 and 500 shares in companies A, B and C respectively.

Company A announced a dividend per share of 3% of the share price, company B paid the investor a dividend of \$60 and company C announced a dividend of 12 cents per share.

- (a) Determine the total dividend the investor can expect from company A if the current price of a share is \$2. (2 marks)

Solution
$2 \times \frac{3}{100} = \0.06 $400 \times 0.06 = \$24$
Specific behaviours
☐ indicates dividend per share ☒ correct dividend

- (b) Calculate the dividend paid per share by company B. (2 marks)

Solution
$\frac{60}{200} = \$0.30 = 30 \text{ cents per share}$
Specific behaviours
☐ indicates calculation ☒ correct amount with units

- (c) Company C did not pay the investor the total dividend but instead paid the balance remaining after withholding a government tax of 20% of the dividend. Calculate the balance due to the investor. (3 marks)

See next page

Solution
$500 \times 0.12 = 60$
$60 \times 0.2 = 12$
$60 - 12 = \$48$
Specific behaviours
☐ indicates full dividend
✓ indicates tax
✓ correct balance

Question 7

(7 marks)

- (a) In a house, the width of a room was 400 cm. On a scale drawing of the house, the same room had a width of 20 mm.

- (i) Determine the scale, in simplest form, used to make the drawing. (2 marks)

Solution
2: 400
1: 200
Specific behaviours
□ writes ratio using same units
✓ simplifies ratio

- (ii) On the scale drawing, a passageway had a length of 45 mm. Determine the length of the passageway in the house, giving your answer in metres. (2 marks)

Solution
$4.5 \times 200 = 900 \text{ cm}$
$900 \text{ cm} = 9 \text{ m}$
Specific behaviours
□ uses scale to determine actual length
✓ converts to metres

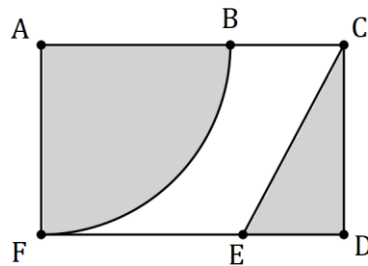
- (b) A plan for a different house was drawn to a scale of 1: 50. A room on the plan measured 60 mm by 80 mm. The floor of the room is to be tiled at a cost of \$40 per square metre. Calculate the cost of the tiling. (3 marks)

Solution
$6 \times 50 = 300 \text{ cm}, 8 \times 50 = 400 \text{ cm}$
$300 \text{ cm} \times 400 \text{ cm} = 3 \times 4 = 12 \text{ m}^2$
$12 \times 40 = \$480$
Specific behaviours
☐ uses scale to determine actual dimensions ✓ calculates area in square metres ✓ correct cost

Question 8

(5 marks)

Rectangular logo $ACDF$, with $AF = 8$ cm and $AC = 12.5$ cm, has a white background with grey triangle CDE and grey sector of circle ABF , as shown below.



Not to scale

Given that $CD = 2 \times ED$ and assuming that $\pi = 3$, determine what percentage of the area of the rectangular logo is grey.

Solution
$ED = 8 \div 2 = 4$
$A_{CED} = \frac{1}{2} \times 4 \times 8 = 16$
$A_{ABF} = \frac{1}{4} \times 3 \times 8^2 = 48$
$A_{ACDF} = 8 \times 12.5 = 100$
$\% \text{ shaded} = \frac{16 + 48}{100} \times 100 = 64\%$
Specific behaviours
✓ length ED □ area of triangle ✓ area of sector ✓ area of rectangle ✓ correct percentage

Supplementary page

Question number: _____

